

Human-to-Robot Handover via Category-Level Shape Fitting and Tactile-Assisted Manipulation

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Abstract—This technical report presents a real-time human-to-robot object handover system developed for the RGM Human-to-Robot Handover Track. The proposed system addresses handover scenarios in which the object is partially occluded by the human hand and where successful grasping depends on both object geometry and hand position. The system combines a dual RGB-D perception pipeline, live-follow robot control, and tactile-assisted grasp and release. In the perception pipeline, two calibrated RGB-D cameras observe the hand-object scene from complementary viewpoints, while a prompt-based object segmentation model extracts object masks from each RGB image. The resulting masks are combined with aligned depth images to lift object regions into per-view 3D point clouds, which are merged, and fitted to category-level templates for stable object geometry estimation under partial occlusion and noisy depth measurements. The fitted object representation is then combined with the estimated human hand state to generate a grasp target on a stable object region away from the hand. The UR5 robot tracks the continuously updated target using a live-follow strategy and executes grasp-and-place motions with a Robotiq 2F-85 gripper equipped with an eFlesh-style tactile fingertip for contact-aware grasping and release. In the competition, the system completed 19 out of 24 trials, achieving an overall success rate of 79.2%. The results show reliable performance for objects covered by the predefined perception and fitting pipeline, while failures were mainly caused by object segmentation failure and invalid grasp target generation.

Index Terms—human-to-robot handover, shape fitting, tactile sensing, robot grasping

I. INTRODUCTION

Human-to-robot handover is a task in which a robot safely and stably receives an object held by a human. In this process, the robot considers not only the object pose but also the position and motion of the human hand. Since the object is often partially occluded by the hand and its size, shape, and orientation can vary over time, reliable grasping cannot be achieved using only 2D detection or a fixed grasp pose. Therefore, the robot estimates the 3D state of the object and the human hand from visual observations, and generates a feasible grasp target in real time while avoiding interference with the human hand.

In this technical report, we present a dual RGB-D vision and tactile-assisted handover system developed for the RGM Human-to-Robot Handover Track. The proposed system observes the hand-object scene from two viewpoints using two Intel RealSense RGB-D cameras. For each view, a prompt-based YOLOE [1] segmentation model estimates the object mask from predefined container-category prompts. Each mask is combined with the corresponding aligned depth image to

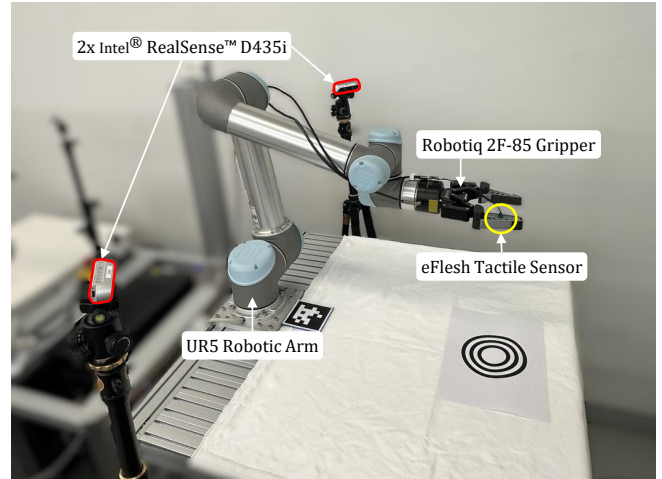


Fig. 1. Hardware setup of the proposed human-to-robot handover system. The setup consists of a UR5 robotic arm equipped with a Robotiq 2F-85 two-finger gripper, an eFlesh tactile sensor mounted on the gripper finger, and two Intel RealSense RGB-D cameras placed on both sides of the robot workspace.

back-project object pixels into a 3D point cloud. The per-view object point clouds are then transformed into the robot base frame and merged into a single object-centric observation. Instead of directly using this raw merged point cloud for grasp planning, the system applies category-level shape fitting to obtain a more stable representation of partial and noisy object geometry. The fitted template provides the object center, scale, orientation, and surface structure, which are used together with the estimated human hand state to generate a hand-aware grasp target.

The robot follows the continuously updated perception result through live-follow control and grasps the object using a Robotiq 2F-85 parallel gripper. In addition, an eFlesh-style tactile sensor attached to the gripper provides auxiliary contact-aware feedback during grasping and release. The final system integrates dual-camera perception, shape fitting-based grasp target generation, robot control, and tactile feedback into a unified end-to-end human-to-robot handover pipeline.

II. HARDWARE SETUP

The hardware setup of our system is shown in Fig. 1. The system is built based on the human-to-robot handover setup introduced in the CORSMAL benchmark [2]. It consists of a 6-DoF UR robotic arm, a Robotiq 2F-85 parallel gripper, two Intel RealSense RGB-D cameras, and a custom tactile

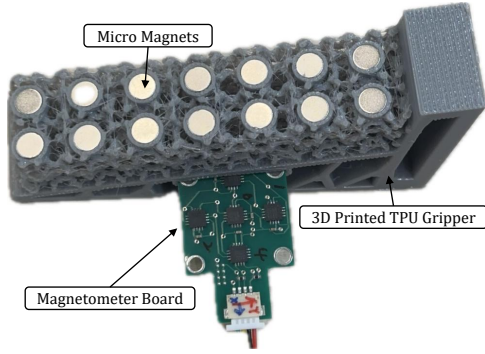


Fig. 2. Custom eFlesh-style tactile fingertip. The TPU fingertip follows the eFlesh [3] microstructured design and contains 14 micro-magnets coupled with a magnetometer board to detect contact-induced magnetic field changes.

fingertip module. The object is initially placed on the table, picked up by the human, naturally handed over to the robot, and then released by the robot at the target location. Two RGB-D cameras were mounted on both sides of the robot workspace to observe the hand-object interaction from complementary viewpoints. The cameras were calibrated with respect to the robot base frame so that visual observations could be transformed into the robot coordinate system. For the end-effector, we used a Robotiq 2F-85 gripper equipped with custom tactile fingertips, as shown in Fig. 2. The tactile fingertips were fabricated using TPU and designed with an eFlesh-style internal microstructure [3]. The right fingertip contains 14 micro-magnets and a magnetometer sensor array, allowing the system to measure contact-induced magnetic field changes during grasping and release. The whole system was executed on a workstation equipped with an Intel Core Ultra 9 CPU and an NVIDIA GeForce RTX 3090 GPU.

III. PROPOSED SYSTEM

The proposed system consists of three main modules: perception, tactile-assisted grasp and release, and robot control. The perception module estimates a hand-aware grasp target from synchronized dual RGB-D observations. The tactile module provides contact feedback during gripper closing and object release, complementing the vision-based target estimation. The robot control module then converts the continuously updated grasp target into executable UR robot motions through a live-follow strategy. Together, these modules allow the robot to approach, grasp, transport, and release a hand-held object in the human-to-robot handover task.

A. Perception Pipeline

The perception pipeline, summarized in Fig. 3, converts synchronized multi-view RGB-D observations into a stable, hand-aware grasp target. For each camera view, the object region is extracted using prompt-based category-aware segmentation and lifted into a 3D point cloud using the aligned depth image. The per-view object point clouds are transformed into the robot base frame and merged to form a unified object observation. A category-level template is then selected, scaled,

and aligned to the merged point cloud, followed by mask-guided scale refinement to improve robustness under noisy or missing depth. In parallel, the human hand state is estimated from dual-view hand landmarks, and the selected hand state is fused with the fitted object representation to generate the final grasp target.

Object segmentation and point cloud lifting. For each camera view, the RGB image is processed by a prompt-based YOLOE segmentation model [1]. We use predefined container-category prompts, including bottle, wine glass, cup, and box-shaped snack, to extract the handover object region from each RGB image. The predicted category c is retained for template selection, and the resulting object mask is combined with the aligned depth image for mask-to-point-cloud lifting. Specifically, valid object pixels inside the mask are back-projected into 3D using the camera intrinsics and depth values. The lifted per-view point clouds are transformed into the robot base frame, filtered to remove noisy depth measurements and sparse outliers, and merged by the object merging module into a single object point cloud.

Category-level shape fitting. Instead of directly using the raw merged point cloud for grasp planning, the system fits a category-level object template to obtain a stable object representation under partial occlusion and noisy RGB-D measurements. The merged object point cloud is first transformed into the robot base frame by the upstream dual-camera pipeline, and the template corresponding to the detected object label c is selected from a predefined template pool.

The initial object scale is estimated from the oriented bounding box (OBB) of the merged point cloud. The canonical template is then scaled along its local OBB axes by matching the size of the template and target extents. This axis-wise scale initialization allows one template to cover category-level shape variations, such as different cup or bottle sizes, while avoiding overfitting to noisy frame-level point cloud details.

Each scaled template is aligned to the merged point cloud using translation-only ICP, and the best candidate is selected according to ICP fitness, RMSE, and the required translation magnitude. The selected scale and rotation are then fixed, so that subsequent frame-to-frame updates do not suffer from unstable rotation changes caused by occlusion or depth noise.

Mask-guided scale refinement. The system additionally performs mask-based scale correction. This step addresses a limitation of RGB-D sensing for transparent objects: while the 2D segmentation mask often provides a reliable object outline, the lifted point cloud can be distorted by depth bleeding or missing depth values. Thus, instead of relying solely on the inaccurate 3D point cloud, the system refines the template scale using the 2D segmentation masks. Nearby scaled template candidates are generated while preserving the ICP-estimated translation and fixed rotation, and each candidate is projected into both RGB-D camera views. The scale is then selected to maximize the multi-view IoU between the projected template and the segmentation masks. This encourages the 3D template to follow the reliable 2D object outline rather than the depth-

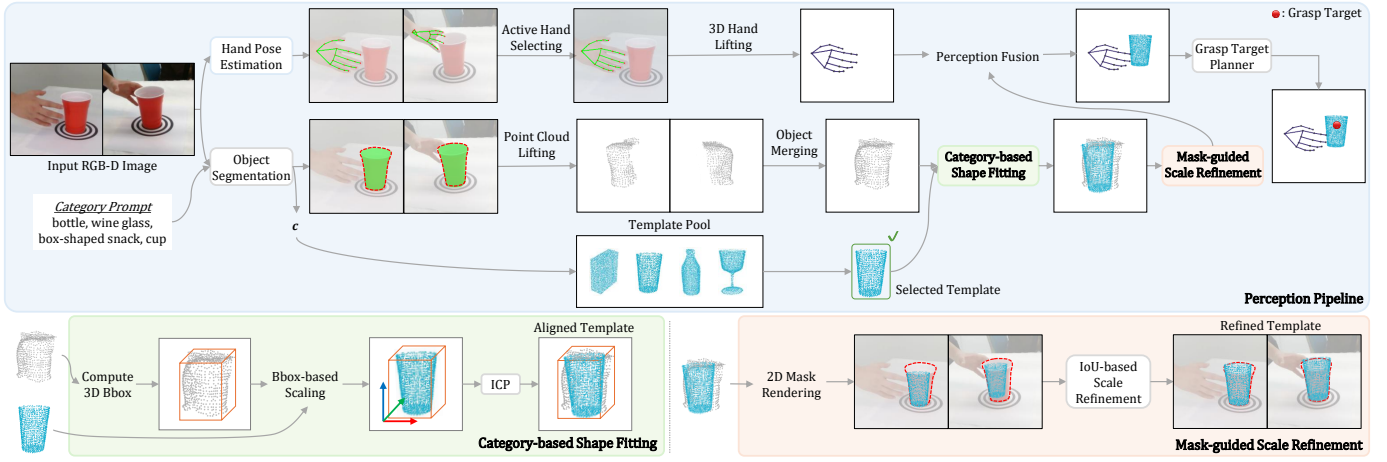


Fig. 3. Overview of the proposed perception pipeline for human-to-robot handover. MediaPipe [4] is used to estimate hand keypoints, while YOLOE [1] performs category-aware object segmentation using predefined category prompts. The segmented object point clouds are lifted into 3D, merged in the robot base frame, and passed to the shape fitting module. A category-specific template is selected from the template pool according to the predicted object class. The selected template is initialized by scaling it to match the 3D bounding box of the merged point cloud and is then aligned to the observed point cloud using translation-only ICP. Finally, the aligned template is projected onto the image plane, and its scale is further refined by maximizing the agreement between the 2D rendered template and the segmentation masks. The refined template is fused with the selected active hand to generate the final grasp target.

distorted point cloud. The final fitted template replaces the raw object point cloud and provides the object centroid, local axes, scale, and surface geometry for downstream hand selection and hand-aware grasp target planning.

Hand state estimation. In parallel with object-centric shape estimation, the system estimates the human hand state from the dual RGB-D observations. We use the MediaPipe hand landmark detector [4] to estimate 2D hand keypoints in each RGB view. The detected landmarks are lifted to 3D using depth values around each keypoint and transformed into the robot base frame. Because the hand can be partially occluded by the object during handover, one camera view may produce unstable 3D hand landmarks after depth lifting. The system performs handedness-based active-view selection. Specifically, in our side-camera setup, right-hand detections use the 3D landmarks from the right-side camera, whereas left-hand detections use those from the left-side camera. The selected landmarks are used to estimate the palm center and palm orientation, preventing unreliable landmarks from the occluded view from affecting the hand-object representation and grasp target planning. The selected hand state is then fused with the fitted object representation to plan a grasp target.

Perception fusion and grasp target planning. The selected hand and fitted object are fused into a temporally stabilized hand-object state. This fused representation encodes the spatial relationship between the object and the human hand and is used by the grasp target planner. The planner searches for a feasible grasp point on the fitted object while rejecting candidates that are too close to the selected hand. As a result, the robot is guided toward a stable object region rather than the region occupied by the grasping human hand. The final output is a continuously updated grasp target, which is sent to the robot controller for live-follow motion. Temporal filtering

Algorithm 1 Tactile-assisted grasp and release.

n_t : current tactile norm; n_r : release reference tactile norm.
 τ_g : grasp-contact threshold; τ_p : placement-contact threshold.

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Move to the planned grasp pose
Start closing the gripper while monitoring tactile norm  $n_t$ 
if  $n_t > \tau_g$  then
    Record the contact gripper position
    Continue closing by a small additional amount
    Stop the gripper
else if gripper position, stall, or force condition is met then
    Stop the gripper
end if
Move to the placement pose
Capture release reference  $n_r$  from the held object tactile signal
Descend the end-effector while keeping the gripper closed
while descending do
    Compute tactile change  $\Delta n_t = |n_t - n_r|$ 
    if  $\Delta n_t > \tau_p$  then
        Stop descent
        Break
    end if
end while
Open the gripper at the final release pose

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and fallback logic are used to reduce discontinuities caused by short segmentation or hand detection dropouts.

B. Tactile-assisted Grasp and Release

In our system, tactile feedback is used as an auxiliary signal during grasping and release. The Robotiq 2F-85 gripper is equipped with an eFlesh-style tactile sensor, which measures magnetic field changes caused by object contact using magnets

and magnetic sensors embedded in the gripper fingers. While the vision module provides the grasp target for approaching the object, the tactile module is used to detect the moment of physical contact during grasping and the contact-state change before release. The overall procedure is summarized in Algorithm 1.

During grasping, once the robot reaches the planned grasp target, the gripper starts closing while monitoring both the gripper state and the tactile response. The tactile signal is represented as the norm of the baseline-subtracted magnetic response vector, denoted by n_t . When n_t exceeds the grasp contact threshold τ_g , the system regards this as object contact. Instead of fully closing the gripper, it records the contact gripper position, performs a small additional closing motion, and then stops the gripper. This allows the robot to secure the object while avoiding excessive closing. If no tactile trigger is detected, the grasp is completed based on the gripper position, stall condition, or force-related state.

During release, the robot moves the grasped object to the placement pose while keeping the gripper closed. Before descending, the system stores a short sample mean of the current tactile norm as the release reference n_r . The end-effector then descends toward the table, and the tactile change $\Delta n_t = |n_t - n_r|$ is monitored. When Δn_t exceeds the release threshold τ_p , the system stops the descent and opens the gripper at the current pose. If no tactile trigger occurs, the gripper is opened when a predefined minimum height or timeout is reached.

In this way, tactile feedback complements vision-based planning by providing contact information that is difficult to observe from vision alone. The vision module determines the grasp and placement targets, while tactile feedback improves execution stability through contact-aware closing and contact-change-aware release.

C. Robot Control

The robot control module converts the continuously updated grasp target from the perception pipeline into executable UR robot motion through the RTDE interface. Since the object can move with the human hand during handover, the system uses a live-follow strategy rather than treating perception as a one-shot planner. The end-effector tracks the latest valid grasp target only when the target satisfies the validity, workspace, and motion-trigger. In addition, the commanded motion is rate-limited at each control cycle to prevent frame-level point cloud jitter or temporary target jumps from producing abrupt robot motion.

During live-follow, the controller updates the Cartesian position (x, y, z) of the end-effector while keeping its orientation fixed at 30Hz. This position-only update provides predictable approach behavior and avoids unnecessary wrist motion during real-time tracking. An approach offset is applied to the grasp target so that the gripper follows a pre-grasp position near the object before executing the final grasp motion.

Once the robot reaches a graspable region, the live-follow controller is paused and the system switches to a structured

TABLE I
COMPETITION RESULTS BY CONTAINER TYPE.

Containers	Success/Trial	Placement error (mm)	Time (s)
Red cup	5/5	13.2	7.2
Plastic wine glass	5/5	15.3	8.1
Chocolate pudding box	4/4	9.6	9.2
Tomato soup can	2/3	17.1	7.7
Frozen food bag	1/3	5	9.9
Stackable food container	2/4	14.2	10.8

Note. Placement error denotes the Euclidean distance between the final object position and the target placement center. Time denotes the elapsed time from program start to the robot's last contact with the object. Both values are averaged over successful trials.

grasp-and-place sequence. The gripper closes at the planned grasp pose, and after grasping, the system stores the relative offset between the end-effector and the tracked object. This object-relative offset is used to move the grasped object back to the delivery area despite small residual errors in the final grasp pose. The robot then moves to a hover pose, descends to the release pose, opens the gripper, retreats upward, and returns to the initial configuration for the next trial.

IV. COMPETITION RESULTS

Table I summarizes the competition results over 24 object-container configurations. Our system completed 19 out of 24 trials, resulting in an overall success rate of 79.2%. The failed configurations were IDs 7, 12, 14, 19, and 21.

A. Results Summary

In addition to the success rate, Table I reports the placement accuracy and task execution time for each container type. The system completed all configurations of the red cup, plastic wine glass, and chocolate pudding box. These objects were covered by the predefined category-level templates, and their segmentation masks were sufficiently stable for point cloud lifting, shape fitting, and grasp target planning. The successful trials also included different contents and fill levels, such as rice, chickpeas, small candy, and empty cases, suggesting that the system was robust to internal content variation when the external object geometry was reliably estimated.

Among the fully successful object types, the chocolate pudding box showed the smallest placement error of 9.6 mm, while the red cup and plastic wine glass showed placement errors of 13.2 mm and 15.3 mm, respectively. The execution time also varied across these objects, ranging from 7.2 s for the red cup to 9.2 s for the chocolate pudding box. These results indicate that even when the handover was completed successfully, the final placement accuracy and task duration were still affected by object geometry, perception stability, target update behavior, and the quality of the estimated grasp and placement pose.

Performance decreased for the tomato soup can, frozen food bag, and stackable food container. The tomato soup can achieved 2 successes out of 3 trials but showed the largest placement error of 17.1 mm. The frozen food bag achieved

only 1 success out of 3 trials, and its small placement error should not be interpreted as reliable placement performance because it was computed from a single successful trial. Overall, these results show that the system performed reliably for objects that were well covered by the predefined perception and fitting pipeline, while performance decreased for objects with unstable segmentation or invalid grasp target generation.

B. Failure Analysis and Discussion

Perception failure. Configurations 12 and 19, both involving the frozen food bag, failed at the object segmentation stage. Our perception pipeline used a prompt-based YOLOE segmentation model with category-level object priors. However, the frozen food bag was not included in the predefined prompt categories, and therefore the object mask was not reliably generated. Since point cloud lifting, shape fitting, and grasp planning all depend on the object mask, this failure prevented the system from generating a valid grasp target.

Grasp target planning failure. Configurations 7, 14, and 21 failed due to invalid or unstable grasp target generation. In configuration 7, the tomato soup can was detected, but the planned grasp target was located too close to the lower boundary of the object. This caused the gripper to contact the object mainly with its fingertip region and push the object out instead of forming a stable grasp. In configurations 14 and 21, which involved the stackable food container, the planned grasp target was generated outside the object boundary.

These failures suggest three main directions for improvement. First, the segmentation module should cover a broader range of container appearances beyond the predefined prompt categories. Second, the grasp planner should verify whether the planned target lies inside the object boundary and away from unstable edge regions. Third, tactile feedback should be extended beyond scalar contact detection to reason about contact location and grasp stability during execution.

V. CONCLUSION AND FUTURE WORK

In this work, we presented a human-to-robot handover system that combines object segmentation, category-level shape fitting, hand-aware grasp target planning, robot control, and tactile-assisted grasp and release. The competition results show that the system can reliably complete handover tasks when the object belongs to a predefined geometric category and the segmentation mask remains stable. However, the failure cases also reveal limitations in category-level perception coverage, RGB-D depth dependence, rule-based grasp target generation, and contact reasoning during execution. Future work will replace category-level template fitting with a category-agnostic shape representation, reduce reliance on RGB-D depth sensing by incorporating stereo RGB and learned 3D priors, and improve grasp target selection through learning-based grasp planning. In addition, tactile feedback will be extended from a grasp/release assistance signal to a placement contact estimation module, enabling table contact-aware release. We also plan to evaluate the system on a broader set of unseen

household objects with more diverse shapes, materials, and contents.

REFERENCES

- [1] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics yolo26: Unified real-time end-to-end vision models," 2026. [Online]. Available: <https://arxiv.org/abs/2606.03748>
- [2] R. Sanchez-Matilla, K. Chatzilygeroudis, A. Modas, N. F. Duarte, A. Xompero, P. Frossard, A. Billard, and A. Cavallaro, "Benchmark for human-to-robot handovers of unseen containers with unknown filling," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1642–1649, 2020.
- [3] V. Pattabiraman, Z. Huang, D. Panozzo, D. Zorin, L. Pinto, and R. Bhirangi, "eflesh: Highly customizable magnetic touch sensing using cut-cell microstructures," 2025. [Online]. Available: <https://arxiv.org/abs/2506.09994>
- [4] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, "Mediapipe: A framework for building perception pipelines," 2019. [Online]. Available: <https://arxiv.org/abs/1906.08172>